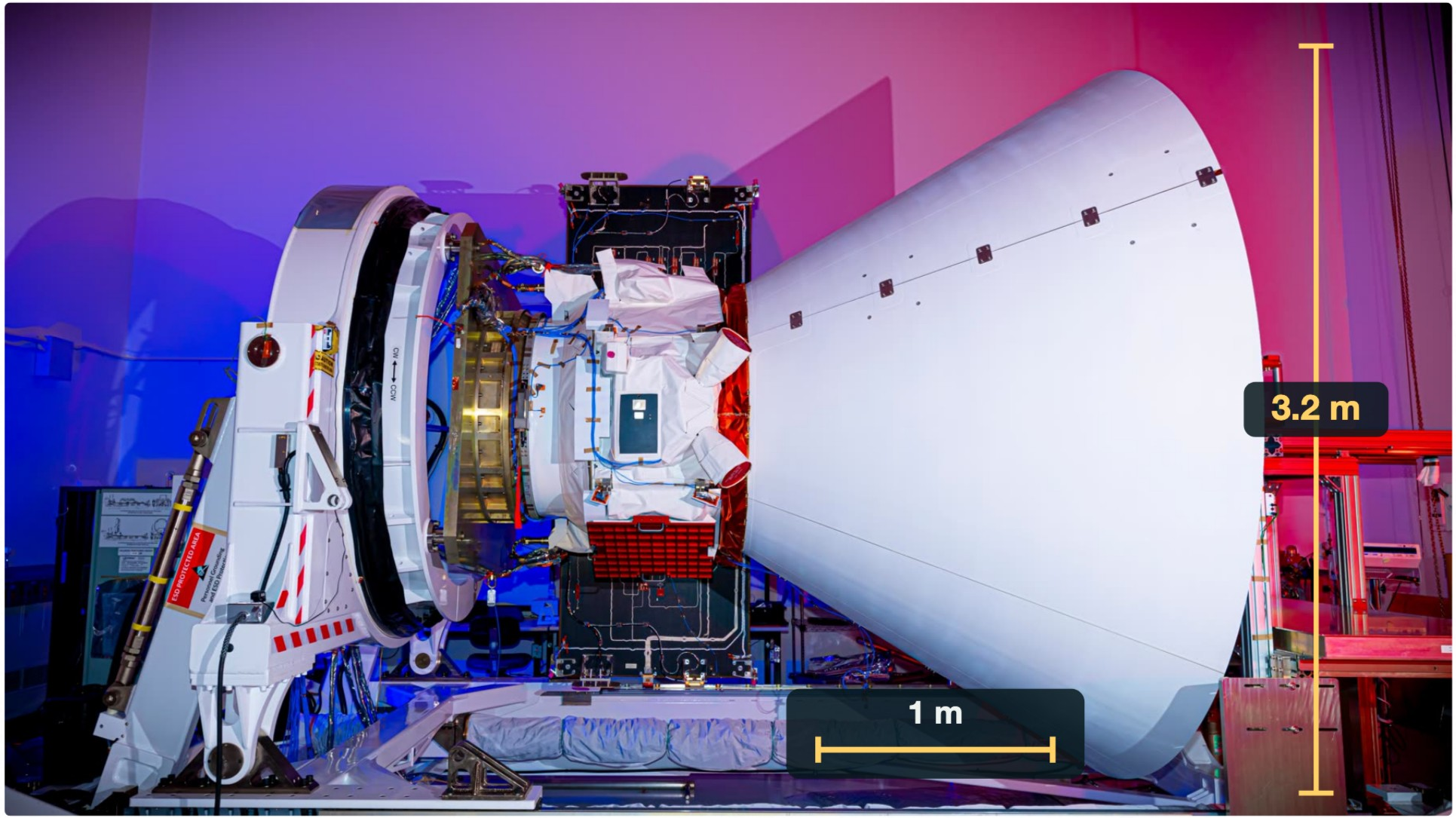
Euclid+SPHEREx

A high purity galaxy catalogue with precise redshifts below $z<0.6$

Calum Murray, Ken Ganga, Martin Kilbinger, Mike Hudson, Cail Daley

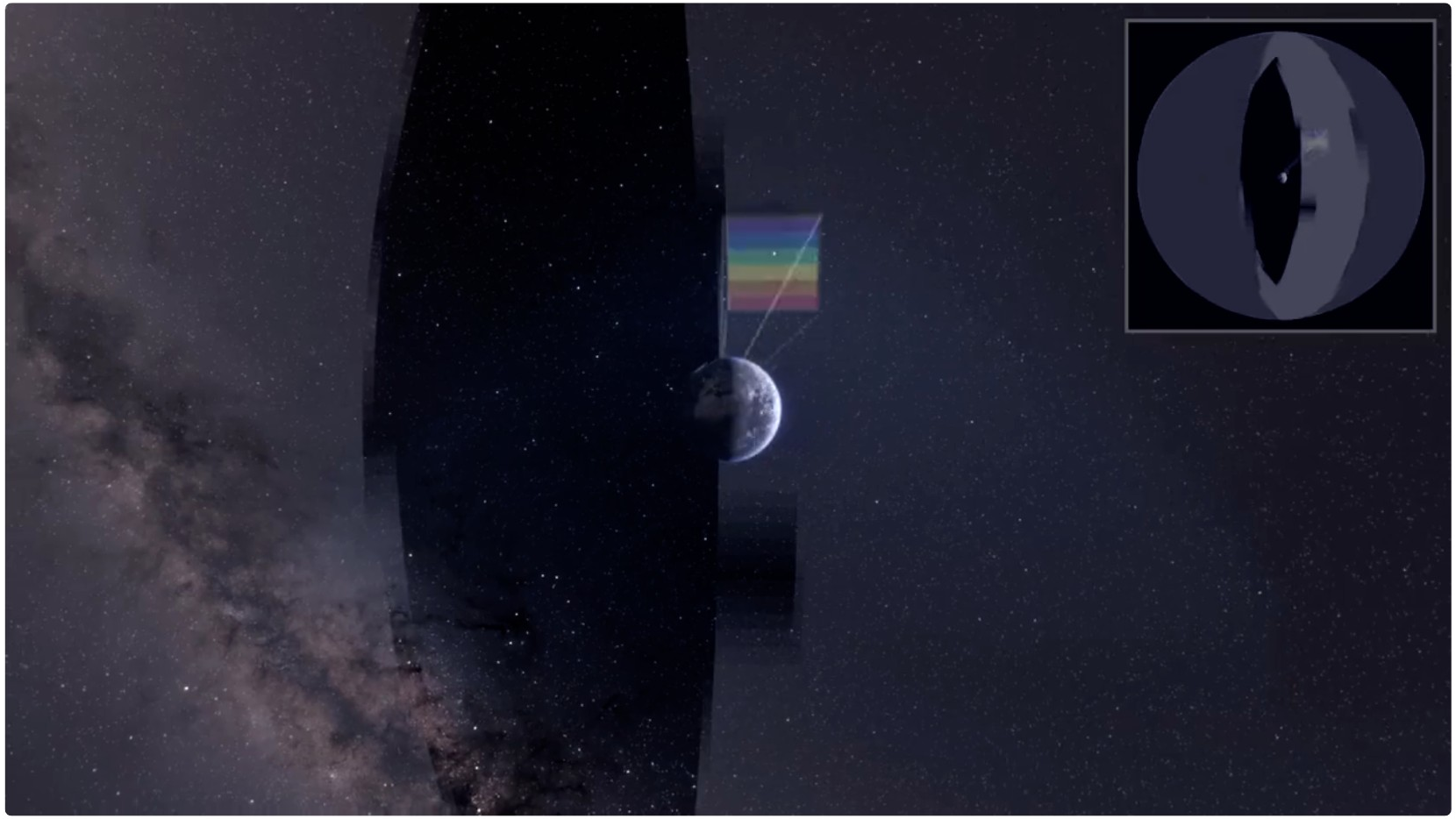
Euclid WL meeting, Napoli · 16 October 2026

The SPHEREx satellite



NASA/JPL-Caltech, PIA26537: SPHEREx after environmental testing, BAE Systems, Boulder, November 2024.

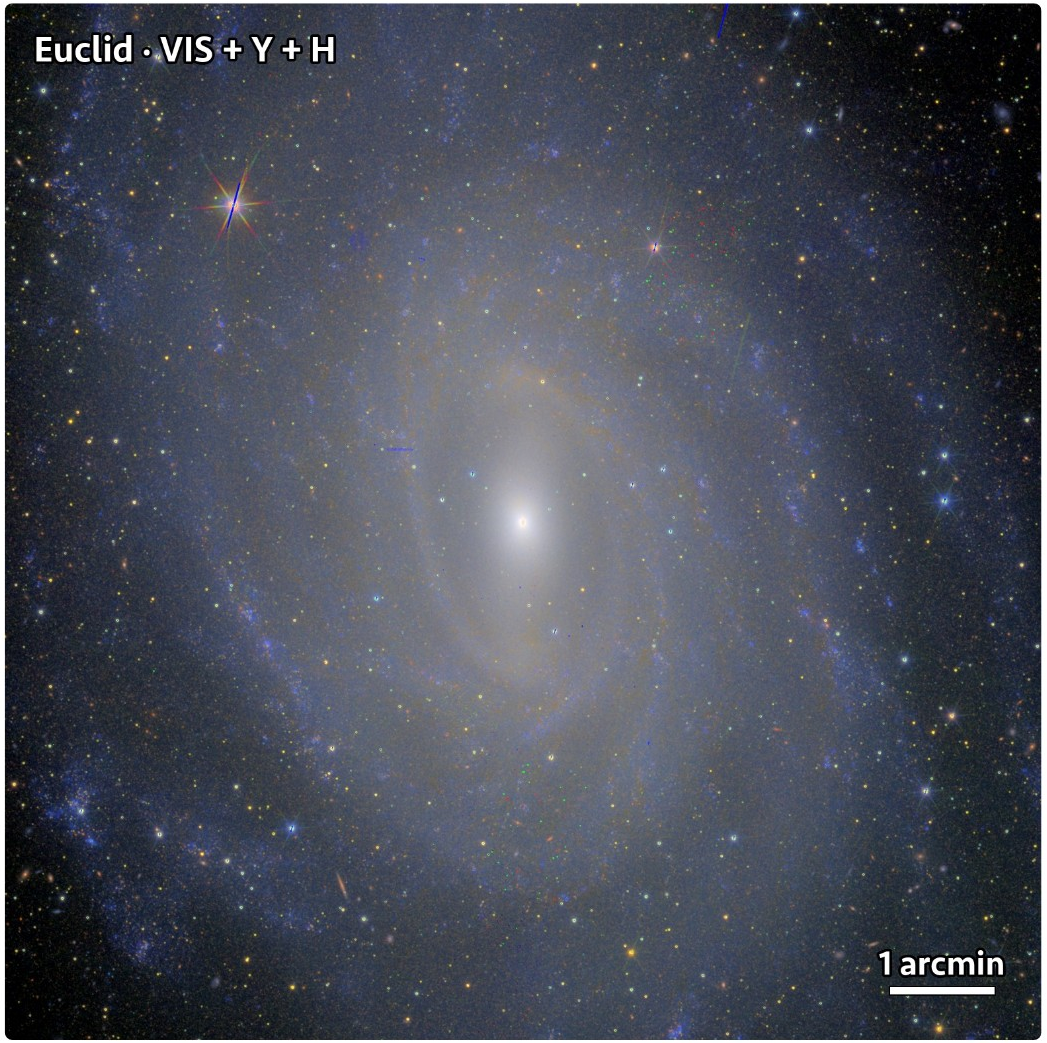
SPHEREx scans the entire sky, at a low earth orbit



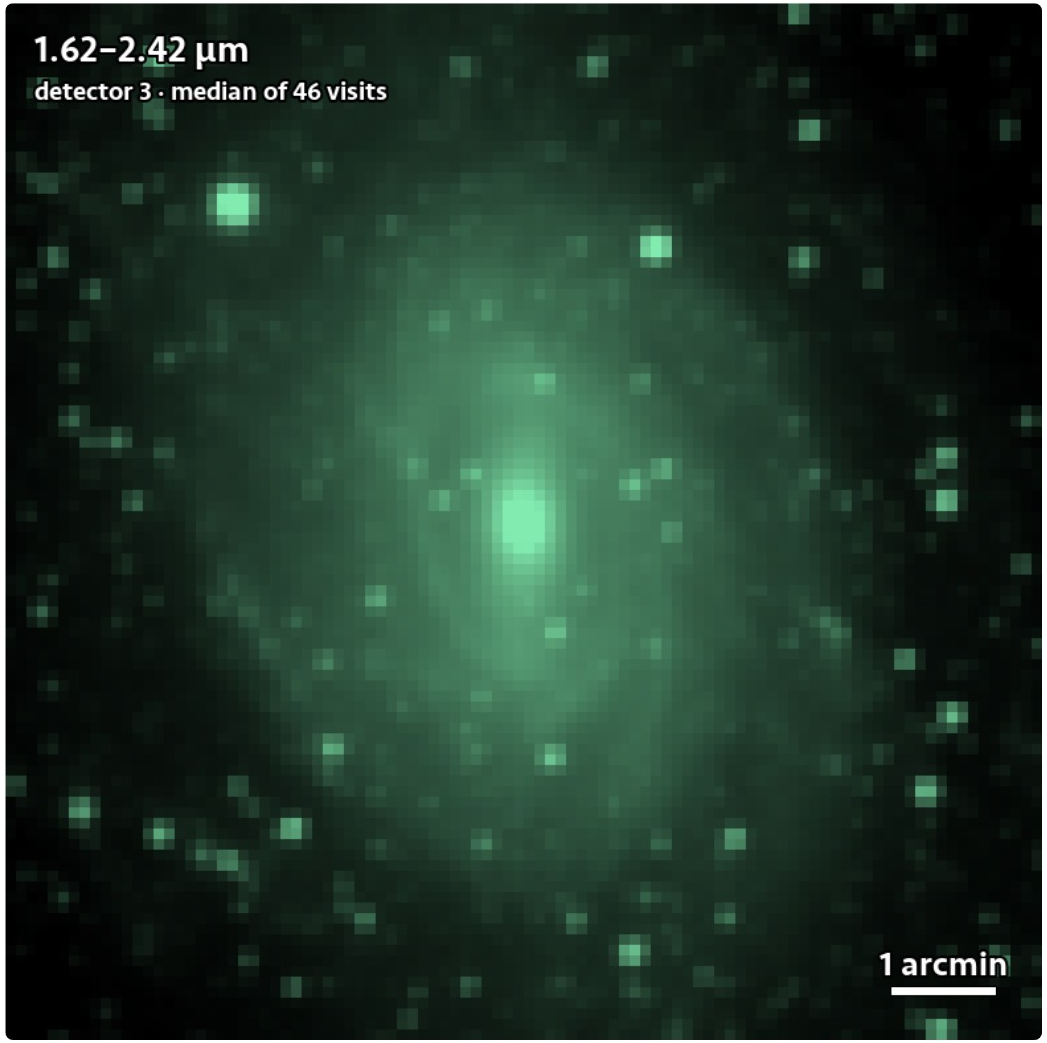
- Launched 11 March 2025; surveying since 1 May 2025
- 20 cm telescope, 3.5° × 11.5° field of view
- Polar orbit, looking ~90° from the Sun: the ecliptic poles are crossed on every orbit
- Orbits ~650 km above the Earth

Animation: NASA/JPL-Caltech (NASA SVS), "Novel Survey Strategy". Numbers: Bock et al. 2025 (arXiv:2511.02985).

A comparison of Euclid and SPHEREx images



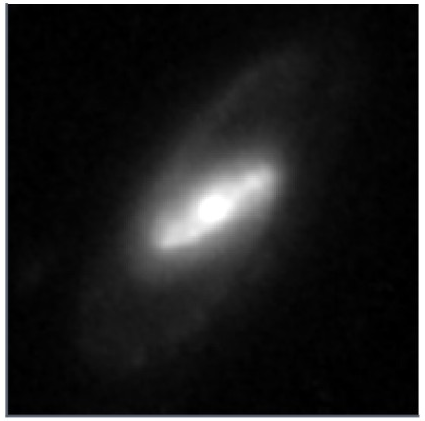
Euclid VIS + Y + H · 0.1″ pixels



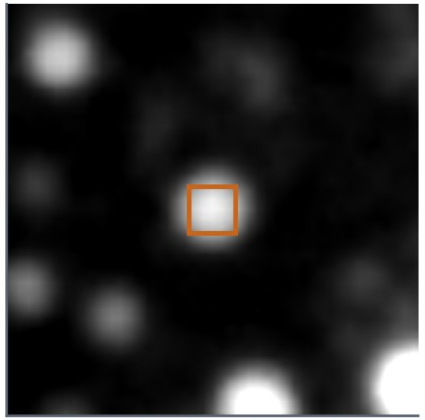
SPHEREx · 6.2″ pixels

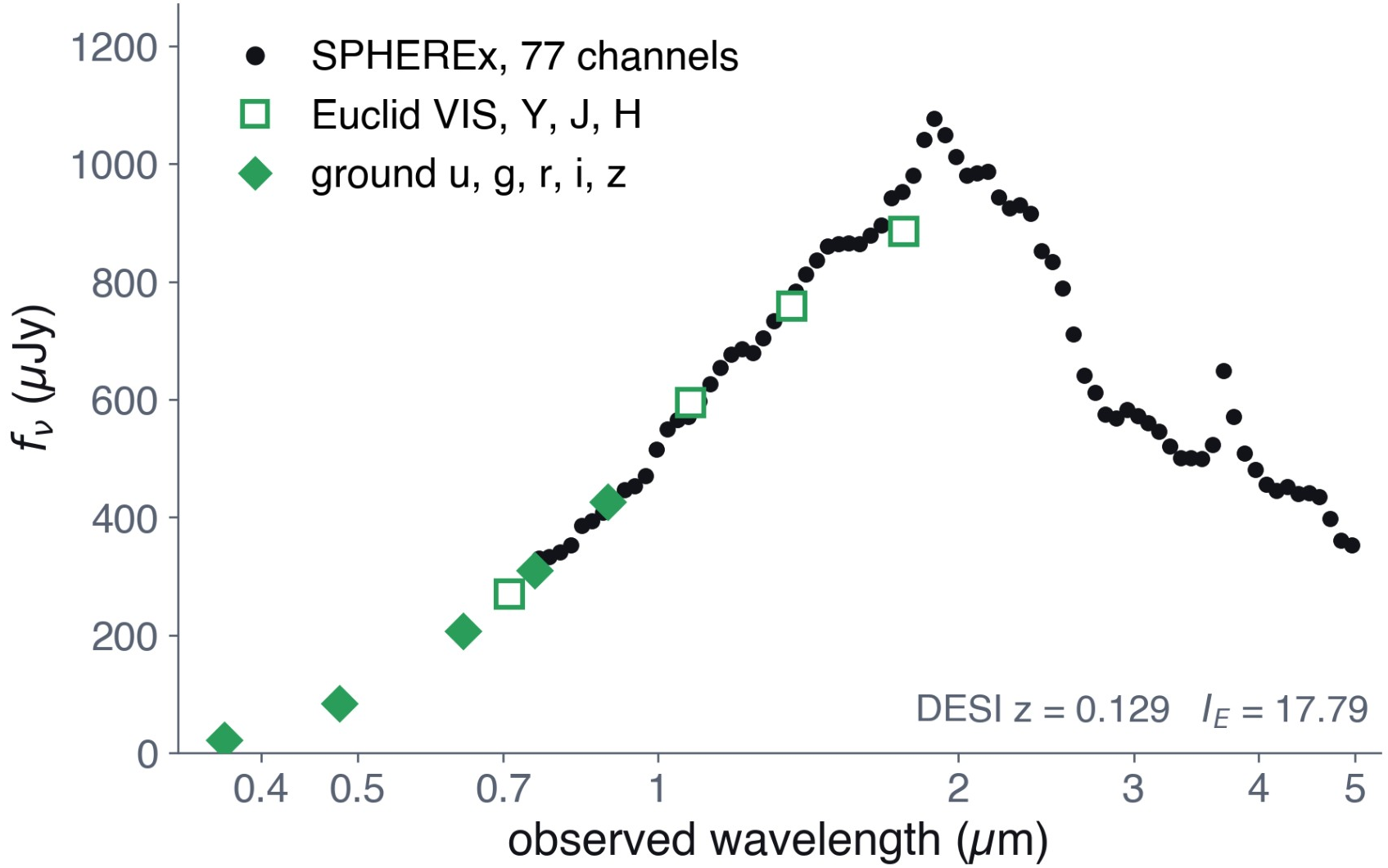
A comparison of SPHEREx and Euclid observations

Euclid VIS · 10″

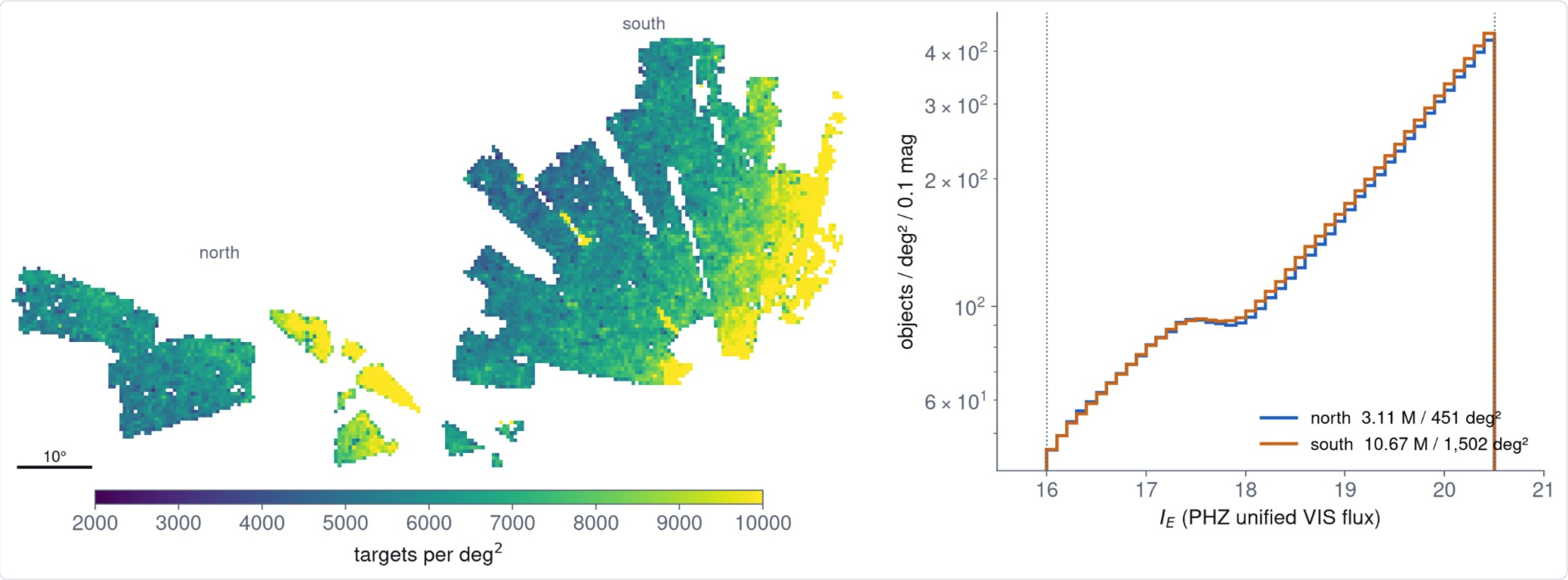


SPHEREx 2 μm · 1.5′

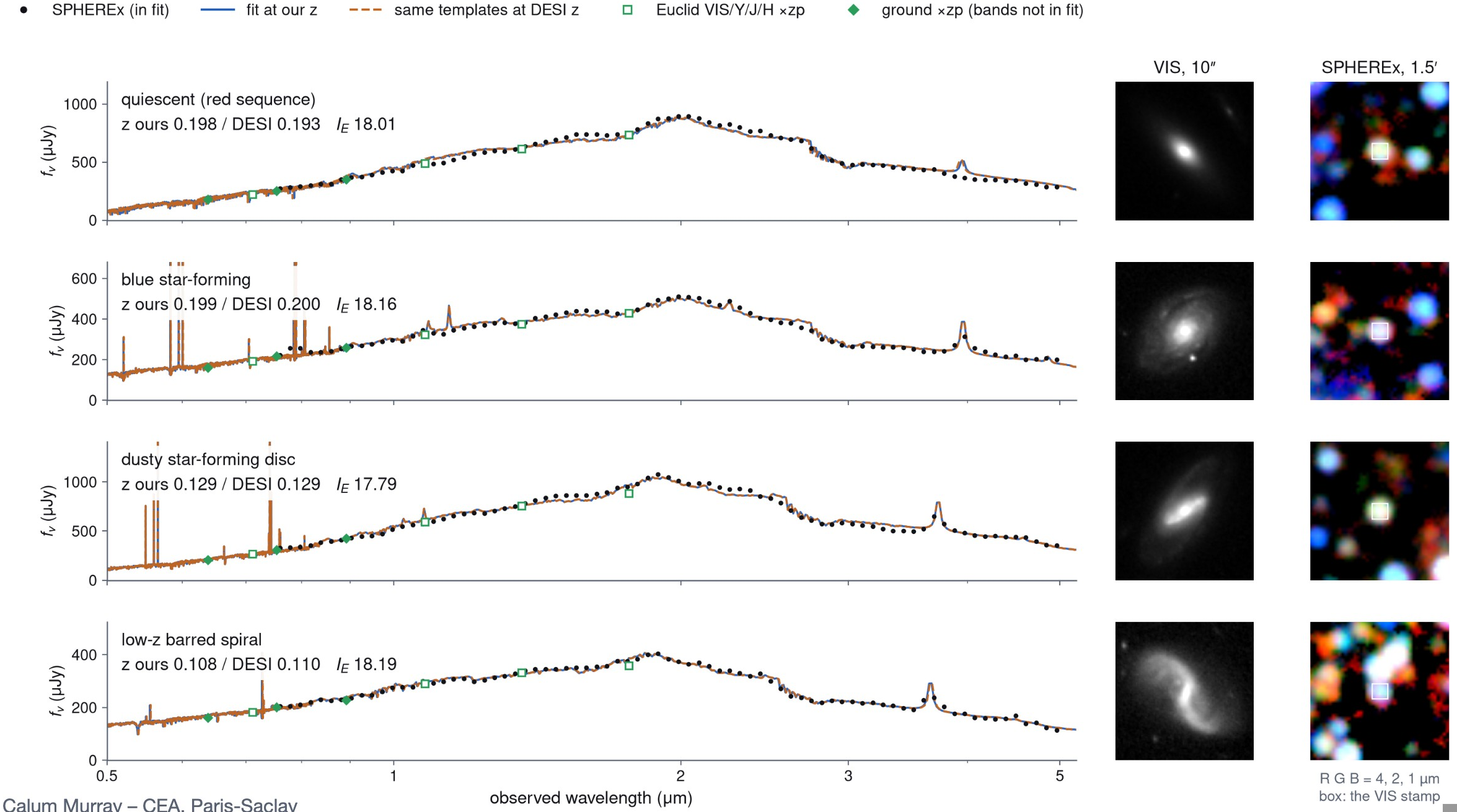




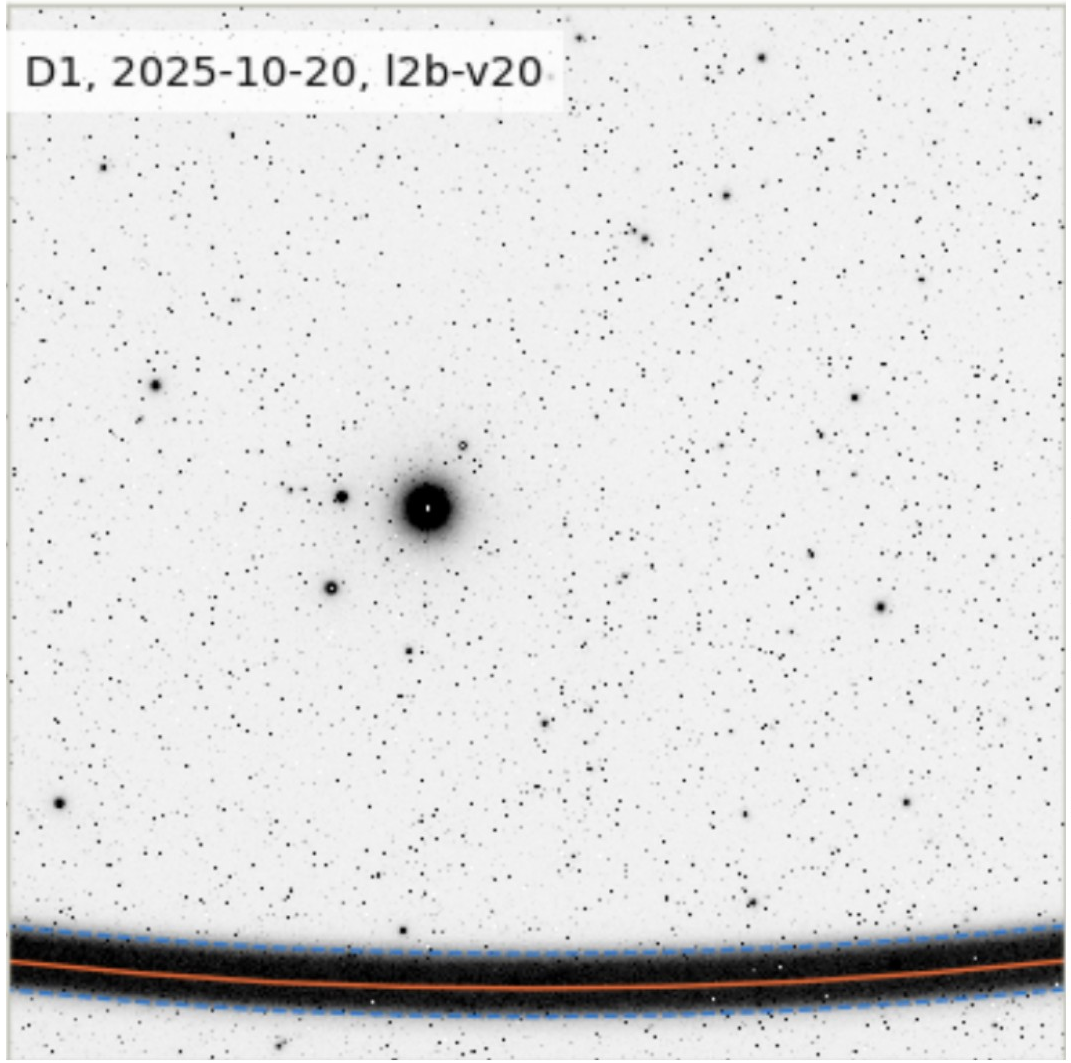
A simple selection of bright VIS sources within Euclid



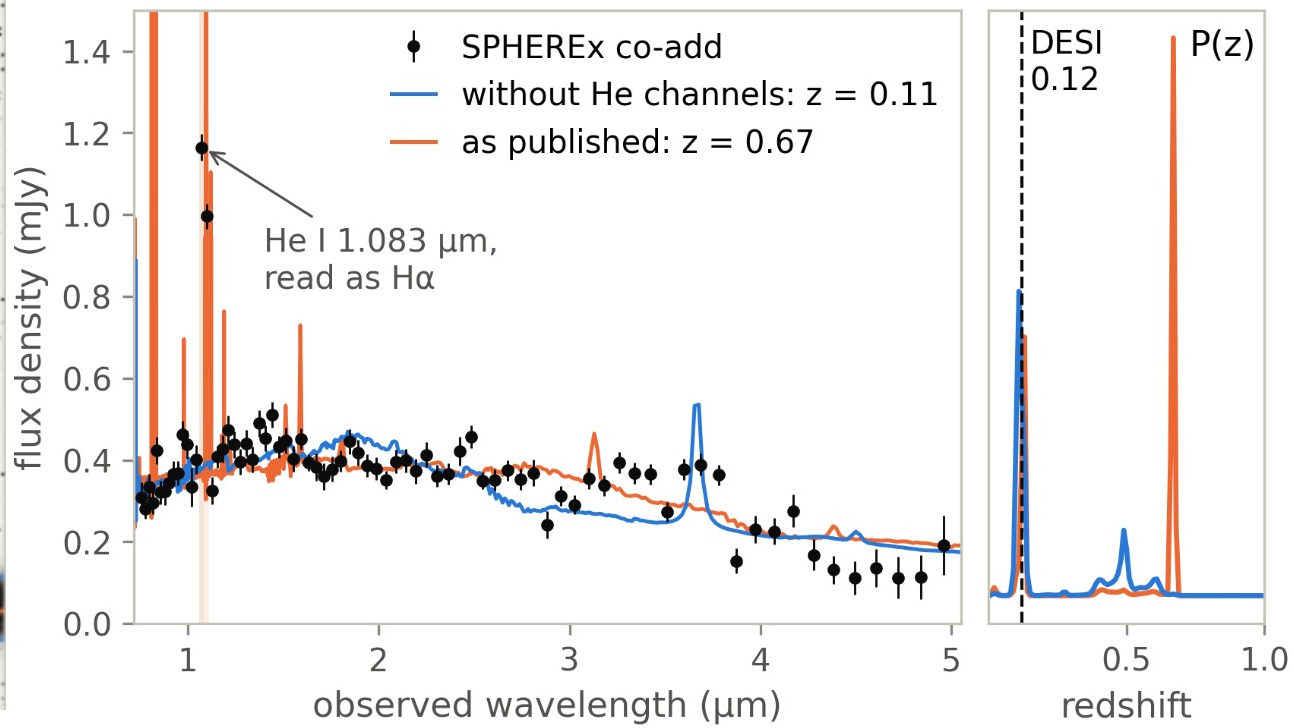
- Select sources with Euclid: 16 < IE < 20.5 on PHZ flux_vis_unif ·
- Peform forced photometry at the position of the Euclid sources within SPHEREx
- Estimate source redshifts with template fitting (Eazy: https://github.com/gbrammer/eazy-py)


SPHEREx (in fit)
fit at our z
same templates at DESI z
Euclid VIS/Y/J/H ×zp
ground ×zp (bands not in fit)
VIS, 10″
SPHEREx, 1.5′
quiescent (red sequence)
z ours 0.198 / DESI 0.193 I_E 18.01
blue star-forming
z ours 0.199 / DESI 0.200 I_E 18.16
dusty star-forming disc
z ours 0.129 / DESI 0.129 I_E 17.79
low-z barred spiral
z ours 0.108 / DESI 0.110 I_E 18.19
f_ν (μJy)
observed wavelength (μm)
R G B = 4, 2, 1 μm
box: the VIS stamp

We can see airglow from Helium with SPHEREx

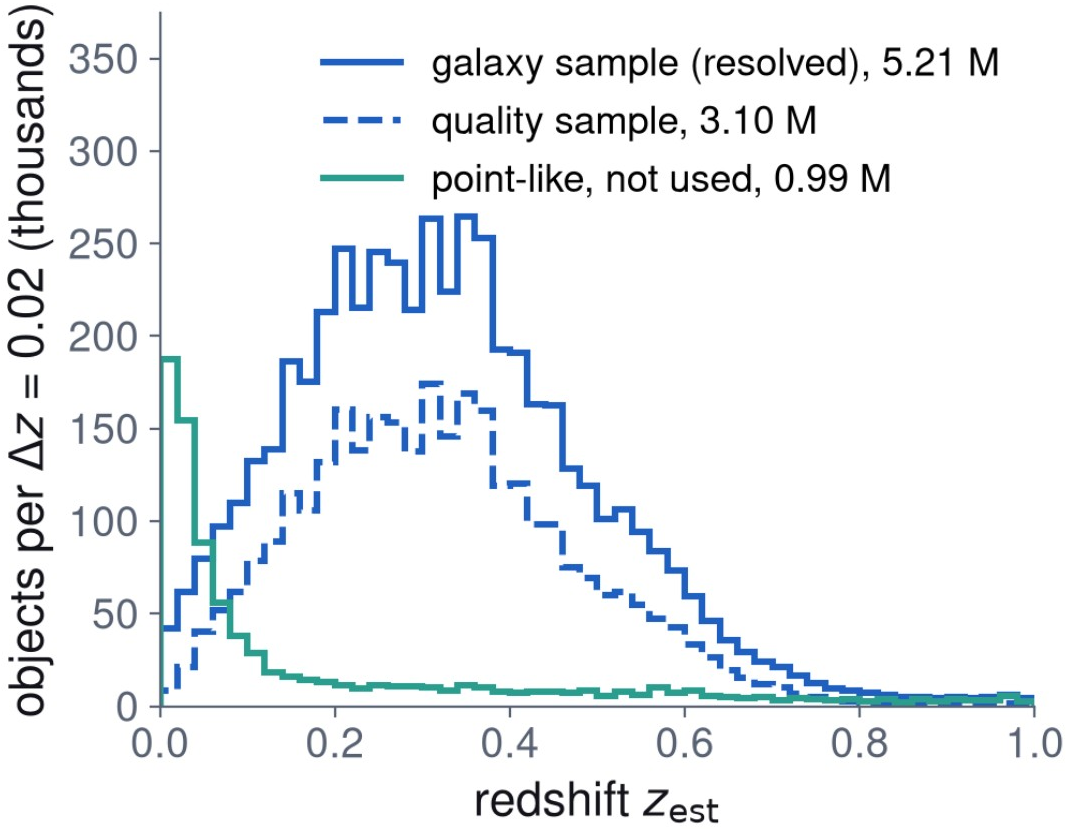


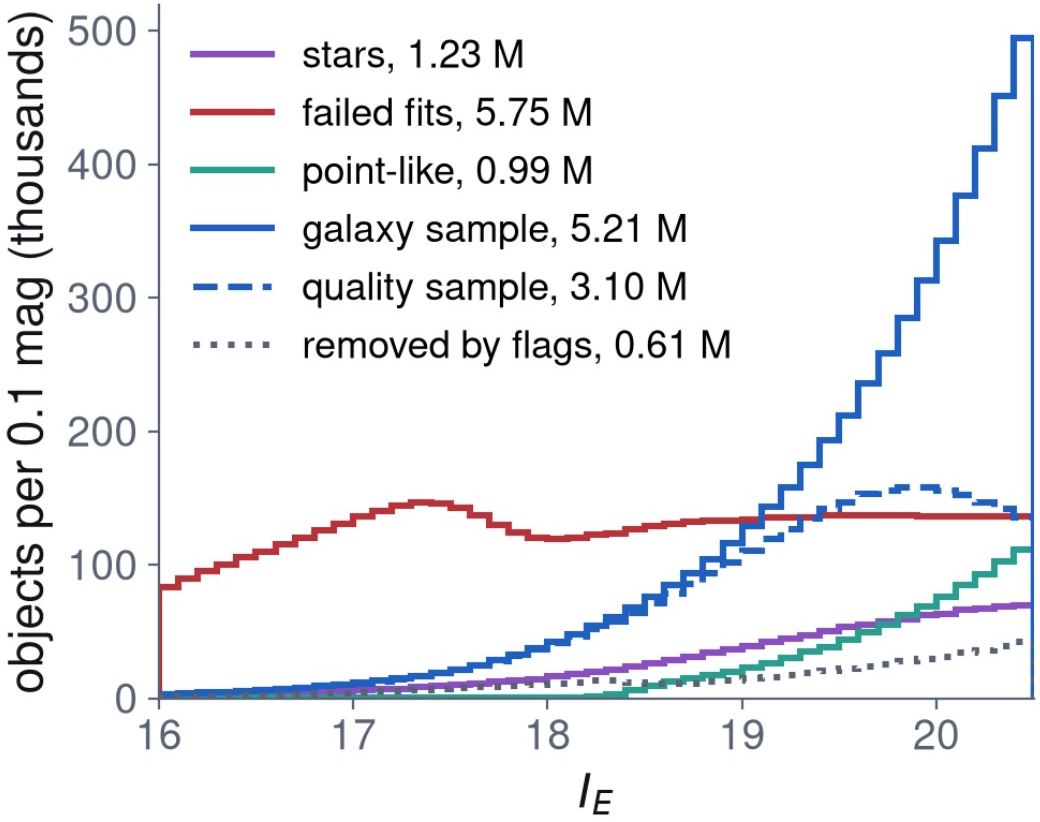
The low-earth orbit means that SPHEREx sees some of the atmosphere.
The sun hits He I in the upper atmosphere.
SPHEREx sees this as emission.
If we're naive it looks like H alpha



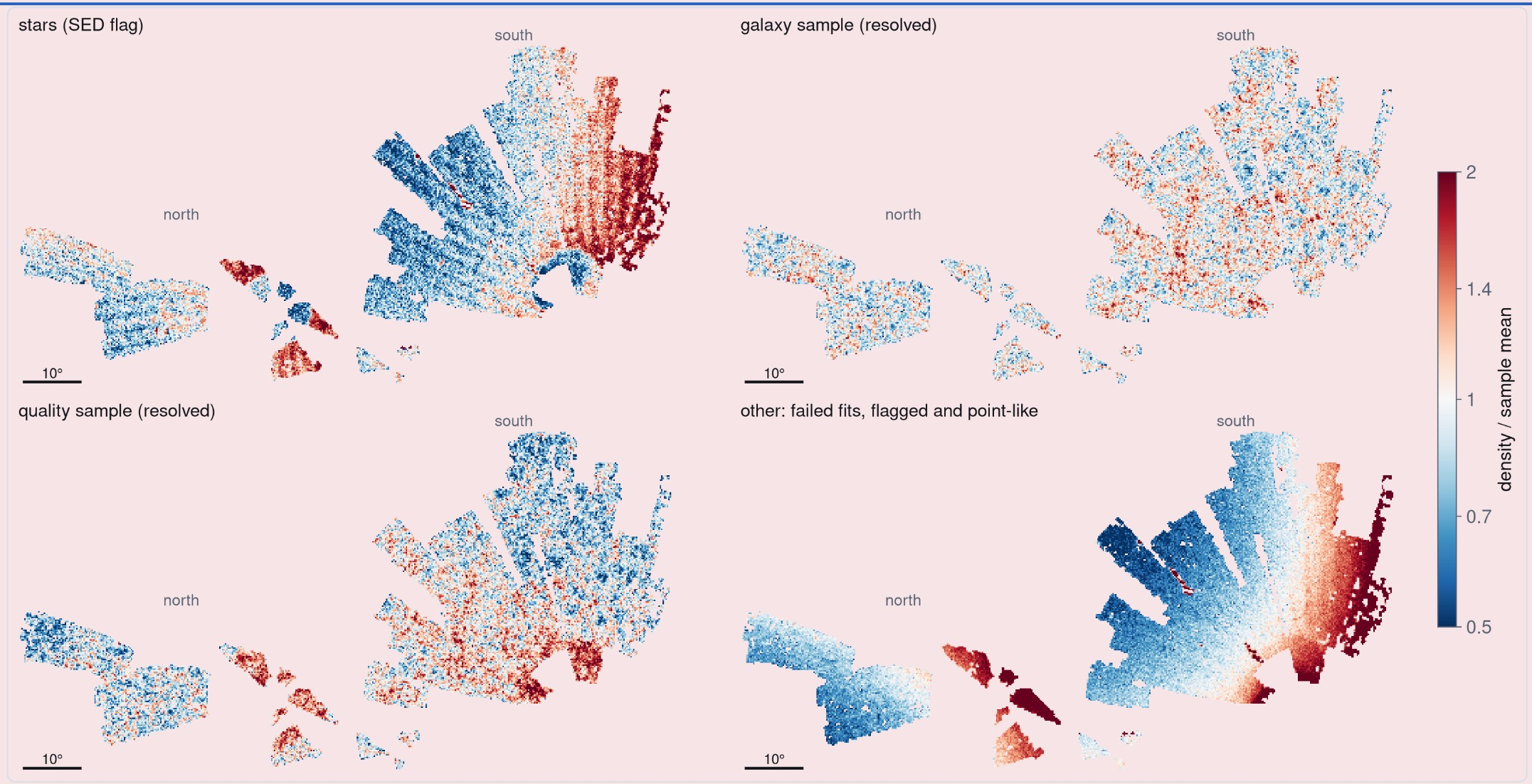
The 13.8 M objects: galaxies, stars and failed fits

ALL OBJECTS	GALAXY SAMPLE	QUALITY SAMPLE	STARS	POINT-LIKE	FAILED FITS	REMOVED BY FLAGS
13,784,773	5,206,897 (37.8%)	3,102,178 (22.5%)	1,233,310 (8.9%)	985,052 (7.1%)	5,751,542 (41.7%)	607,972 (4.4%)





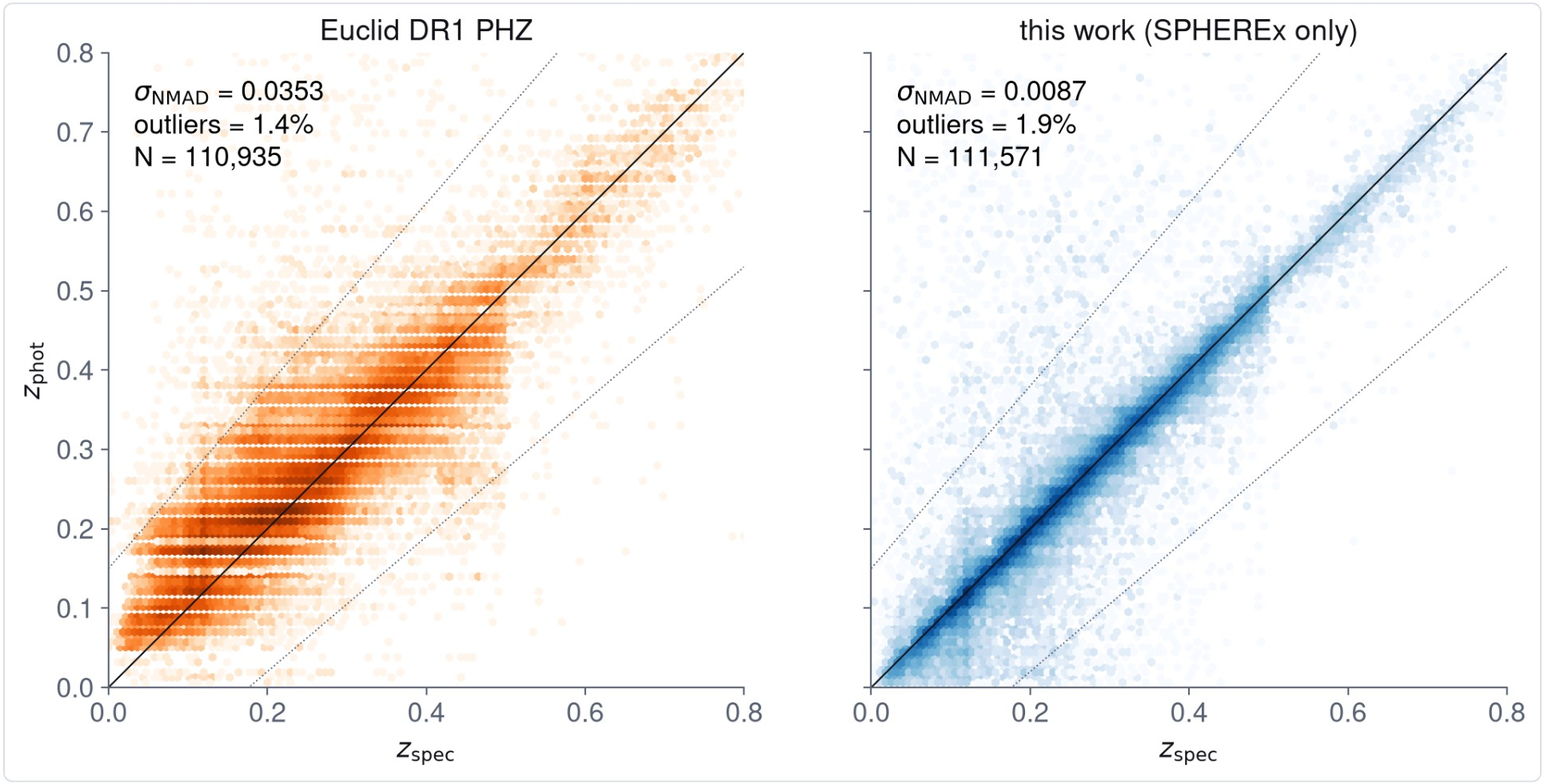
Sample homogeniety



Scrolling through redshift, both lobes

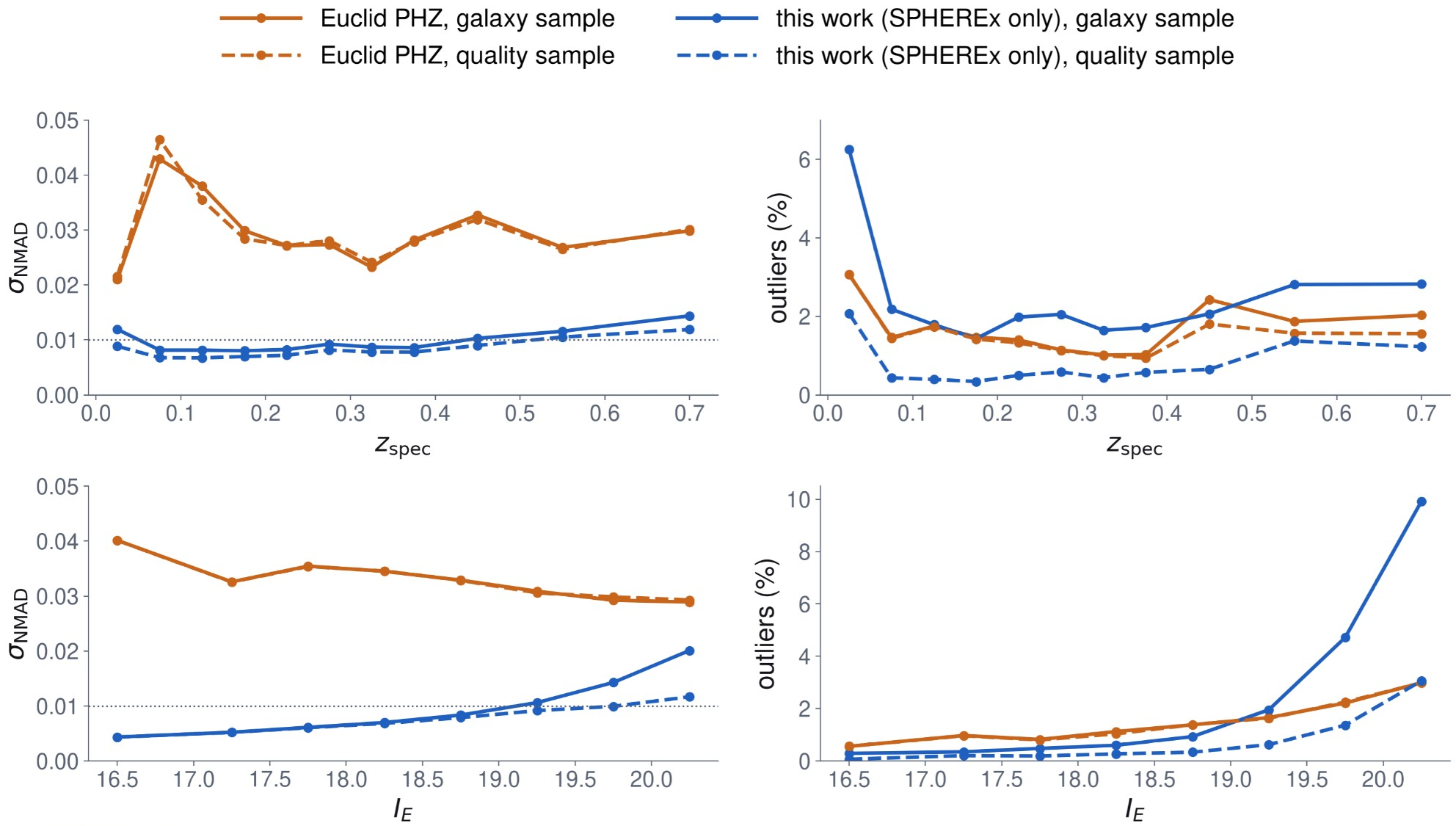


Redshift validation



Truth: DESI DR1 + 6dFGS + 2dFGRS + 2dFLenS + OzDES.

Redshift validation



Galaxy correlations

Eight sky maps: dust, star density, SPHEREx channel count, ecliptic latitude, zodi, background, visits, Euclid VIS depth; five are regressed.

In each shell, galaxy density is binned against each map. A trend counts only if it is 2× larger than in 400 lognormal mocks, which cluster but are clean.

Remove the worst trend with a linear fit, repeat until all pass, and weight each galaxy by 1/fit in its own shell.

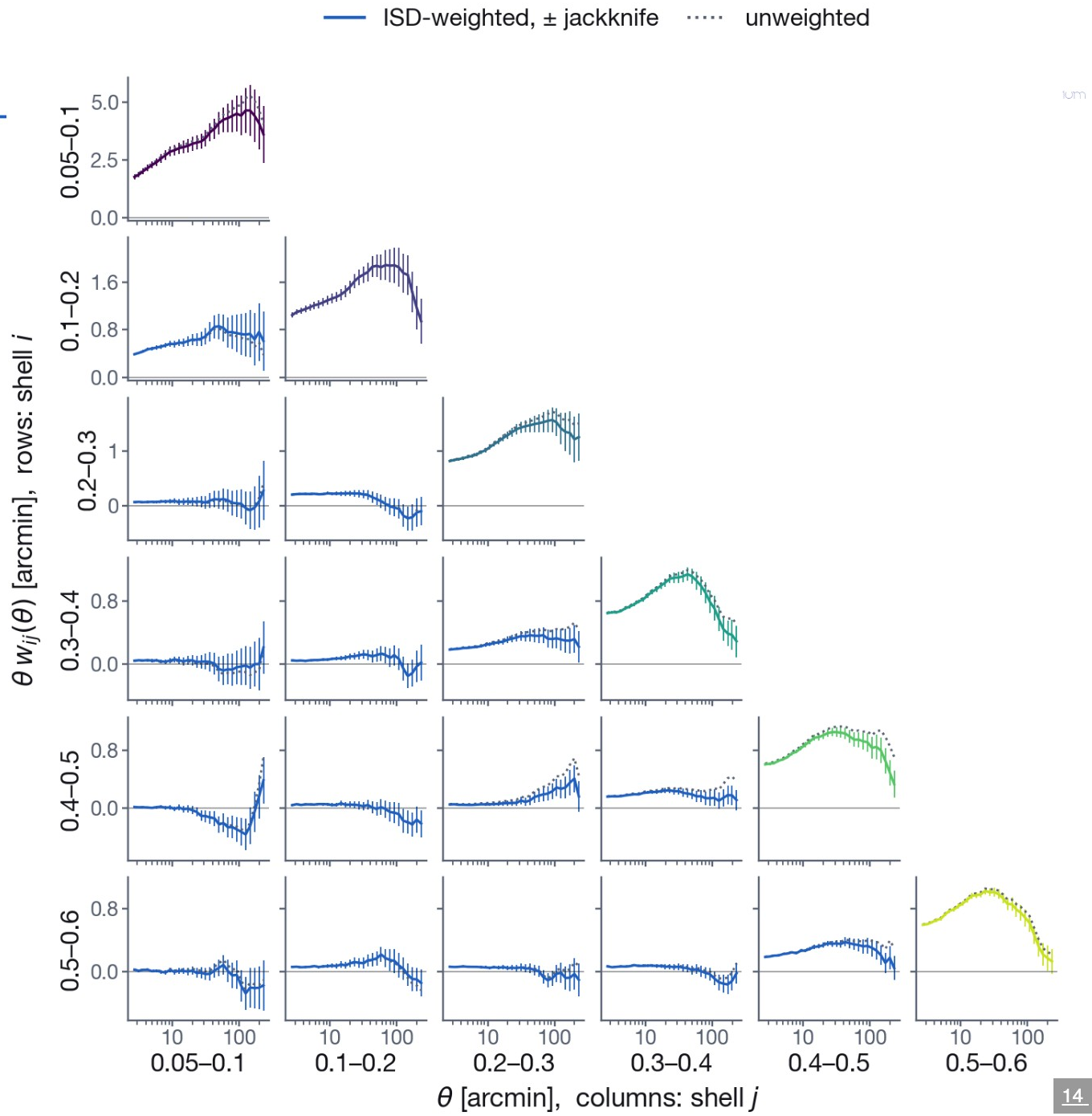
What it shows

Neighbouring shells correlate (r = 0.15–0.33, the photo-z overlap); shells two or more apart are near zero.

The worst trends, removed in order

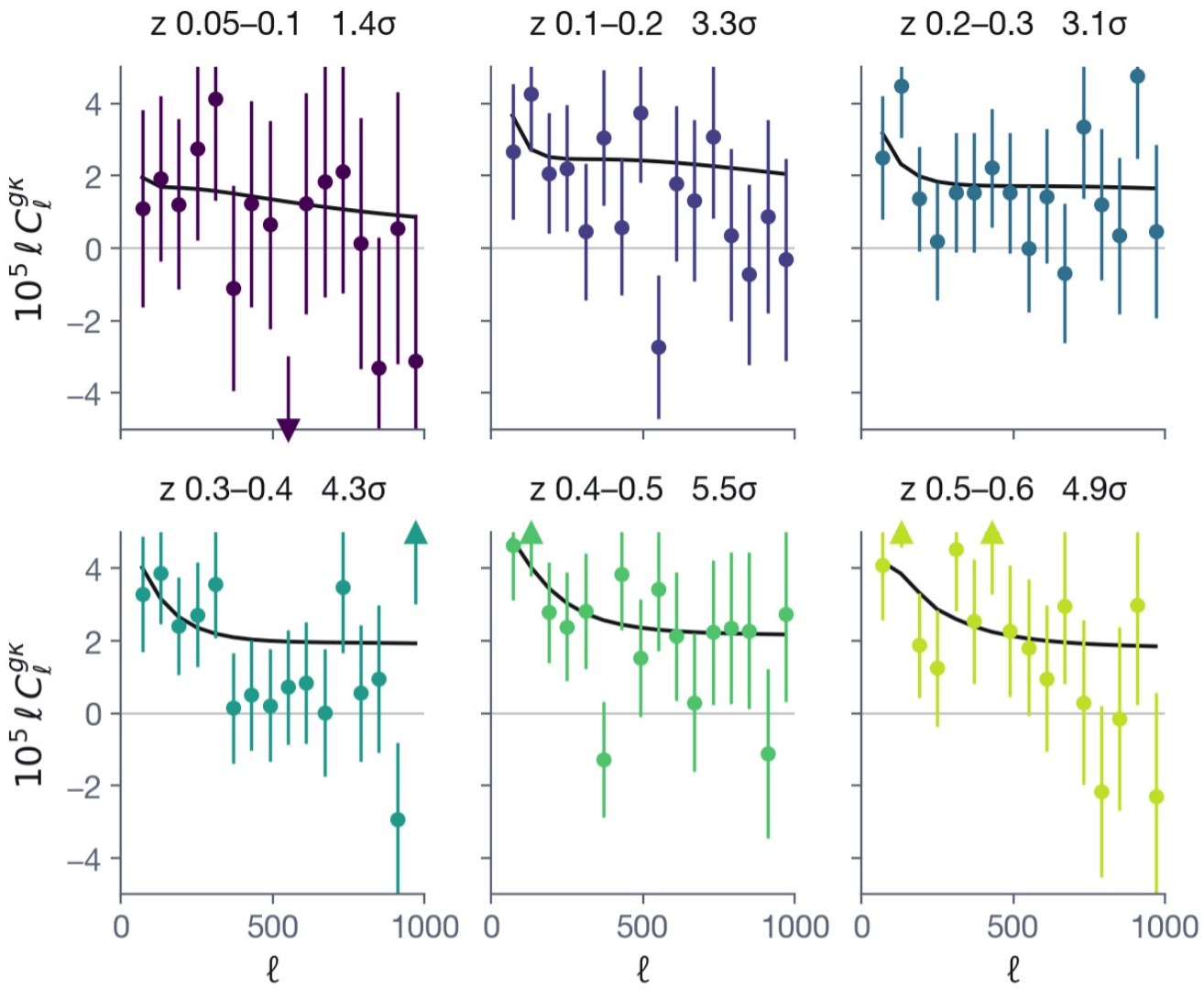
"Worst" = the map whose linear fit gains the most χ^2 over the mocks.

0.05–0.1: stars
0.1–0.2: none
0.2–0.3: channel count, ecliptic latitude, VIS depth
0.3–0.4: channel count, stars, VIS depth
0.4–0.5: channel count, stars
0.5–0.6: ecliptic latitude, channel count
Weights stay within 0.77–1.36.



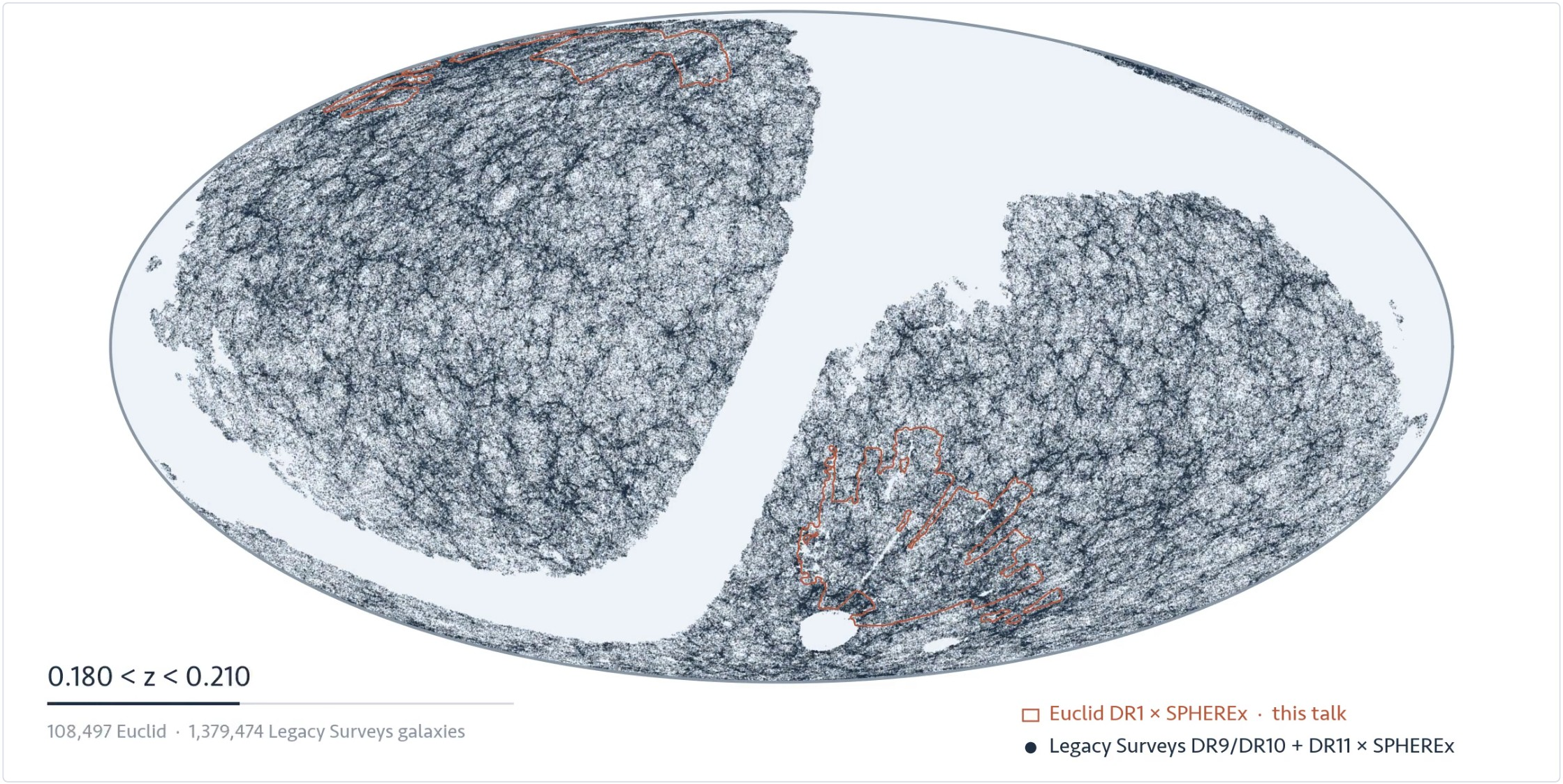
Calum Murray – CEA, Paris-Saclay

CMB lensing cross-correlations



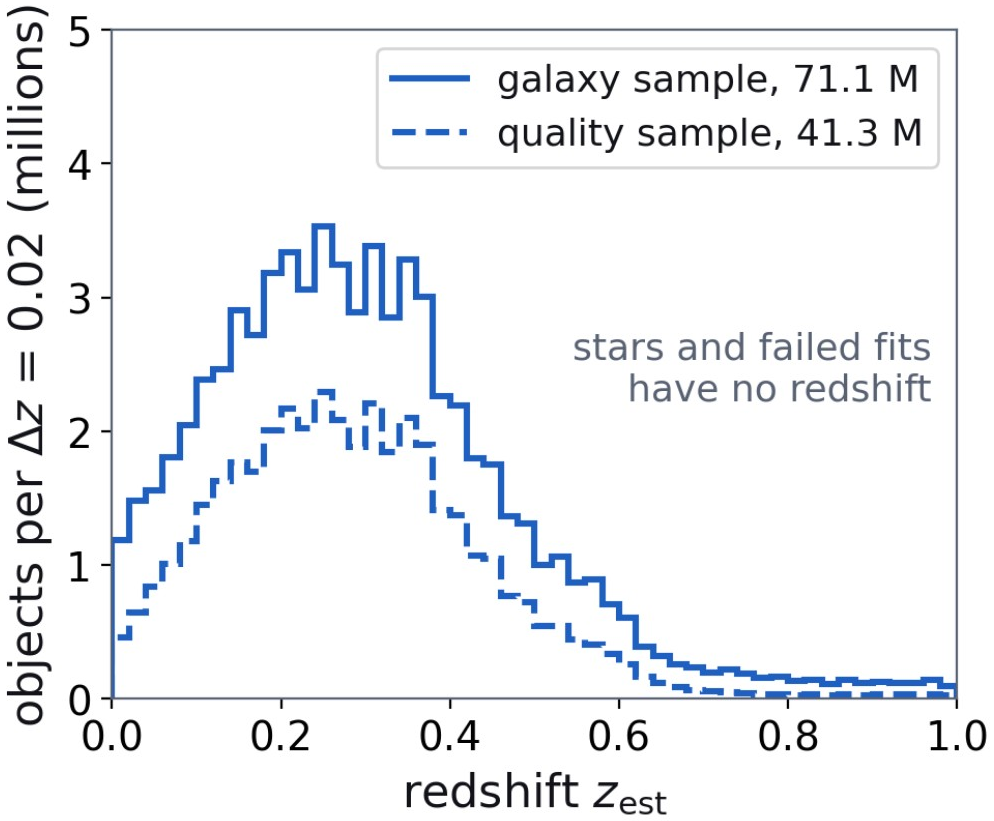
ISD-weighted. Pseudo-Cℓ with NaMaster (Alonso, Sanchez & Slosar 2019, MNRAS).

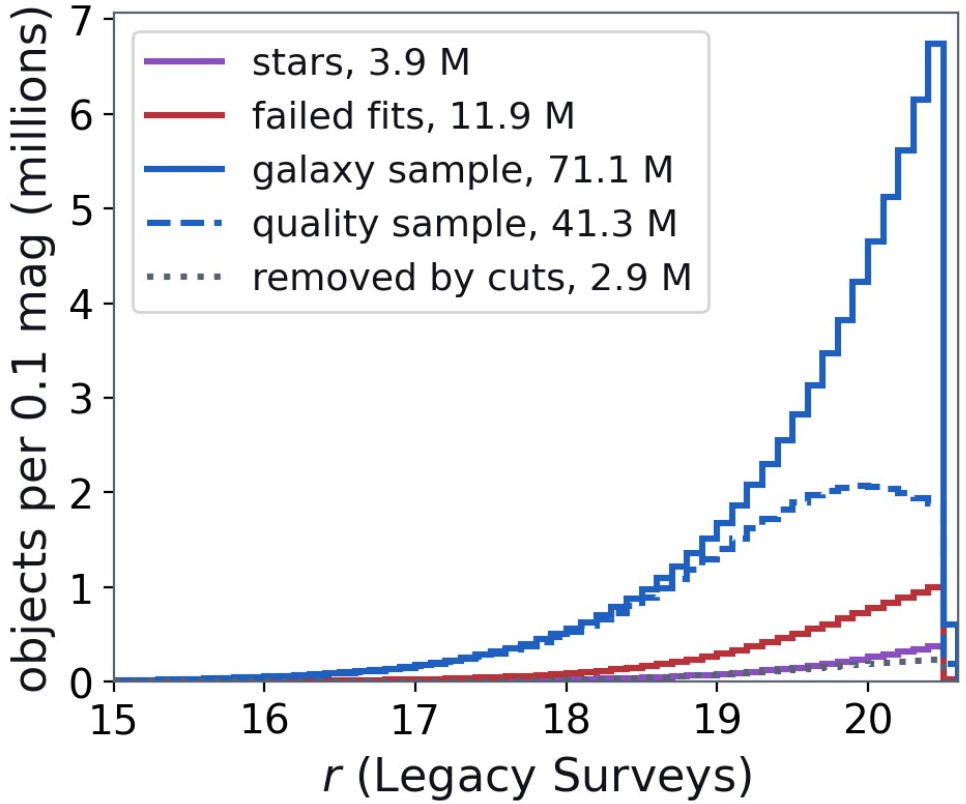
A similar catalogue, but with DESI legacy imaging instead



Some numbers for the full sky catalogue

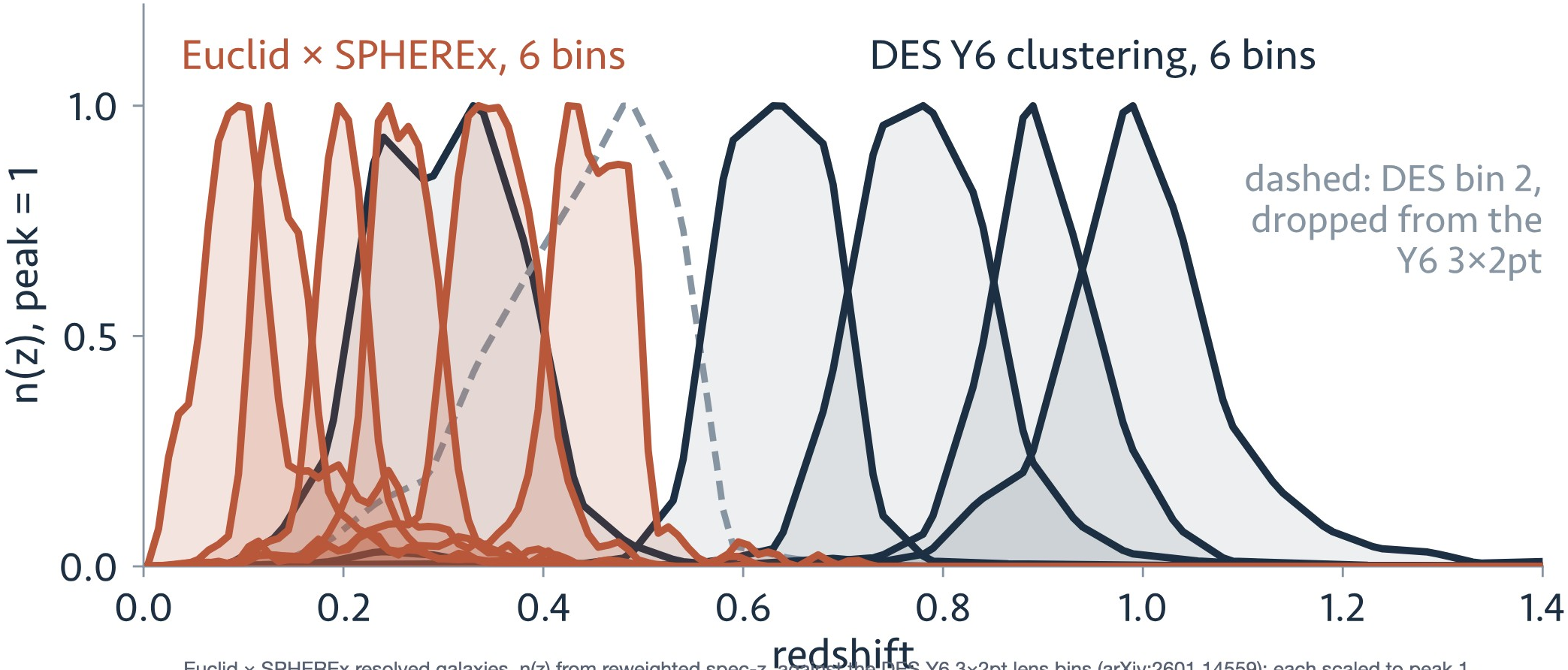
ALL OBJECTS	GALAXY SAMPLE	QUALITY SAMPLE	STARS	FAILED FITS	REMOVED BY CUTS	CALIBRATION STARS
92,036,397	71,137,758 (77.3%)	41,294,489 (44.9%)	3,916,929 (4.3%)	11,872,782 (12.9%)	2,942,193 (3.2%)	2,166,735 (2.4%)





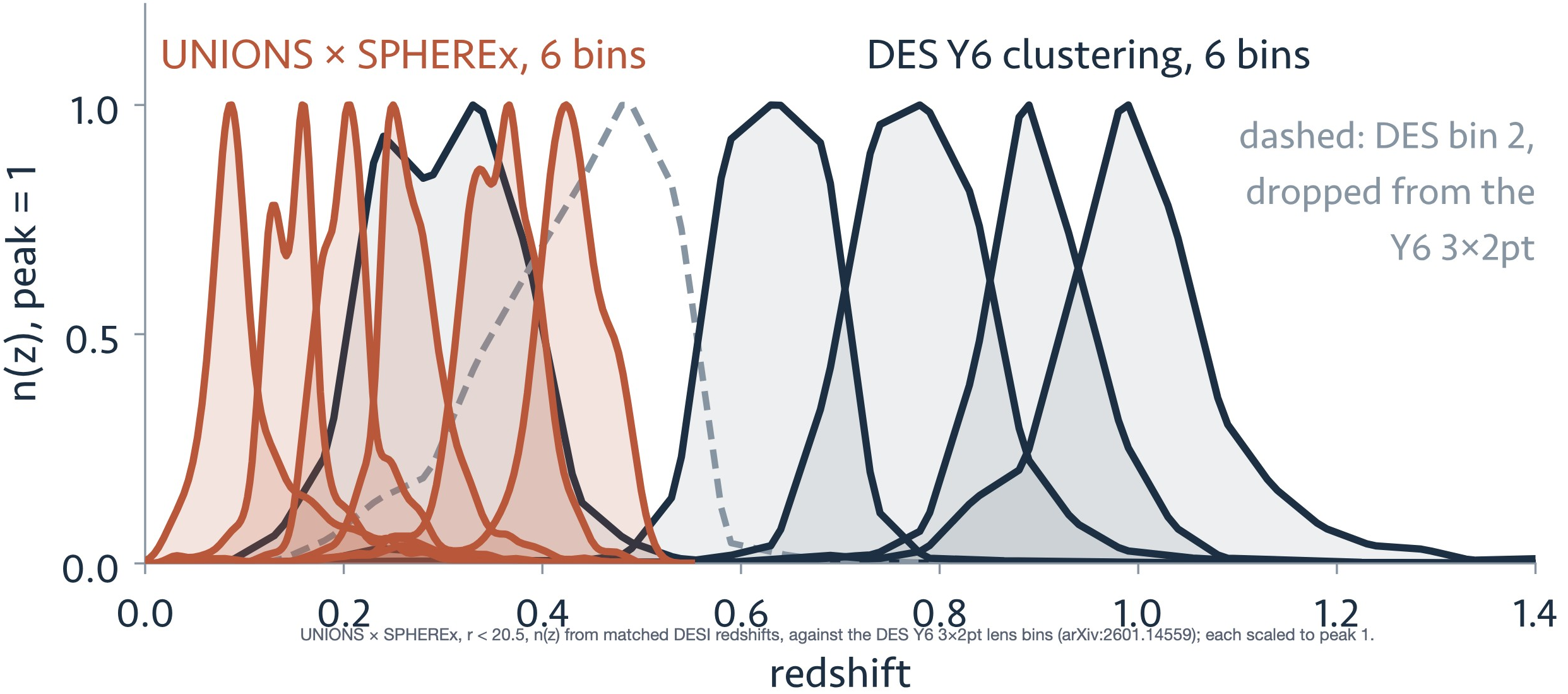
Summary

- A close to final Euclid+SPHEREx catalogue is available for anyone to use!
- The SPHEREx + DESI legacy imaging surveys sample is also available for anyone to use!
- If you're interested send me a message



Euclid × SPHEREx resolved galaxies, n(z) from reweighted spec-z, against the DES Y6 3×2pt lens bins (arXiv:2601.14559); each scaled to peak 1.

The same comparison for UNIONS × SPHEREx

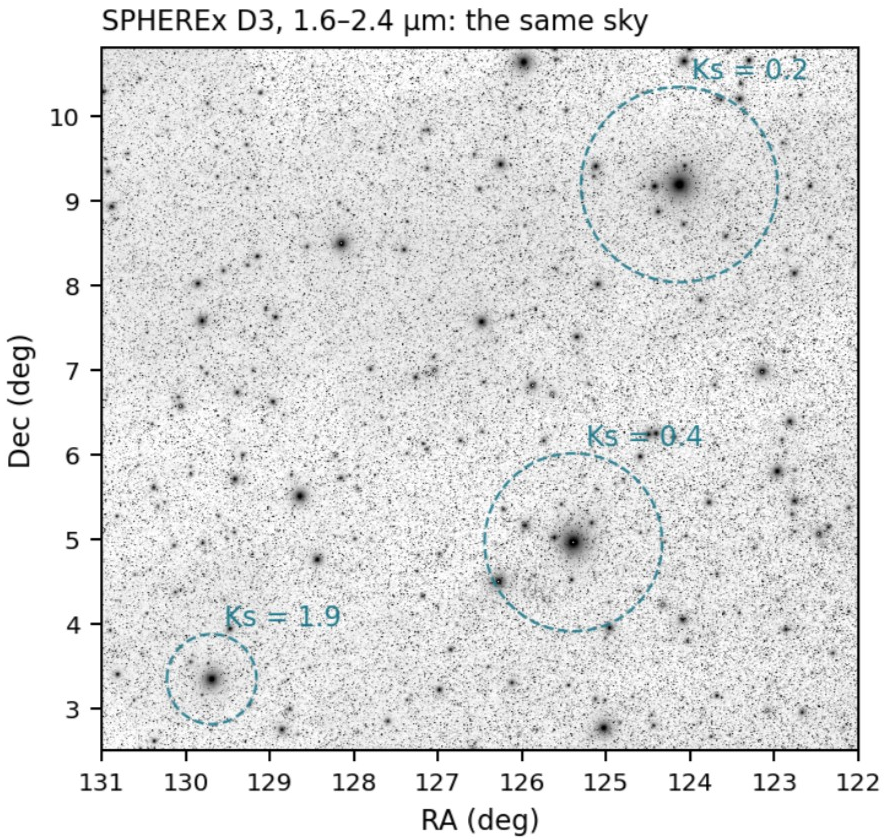


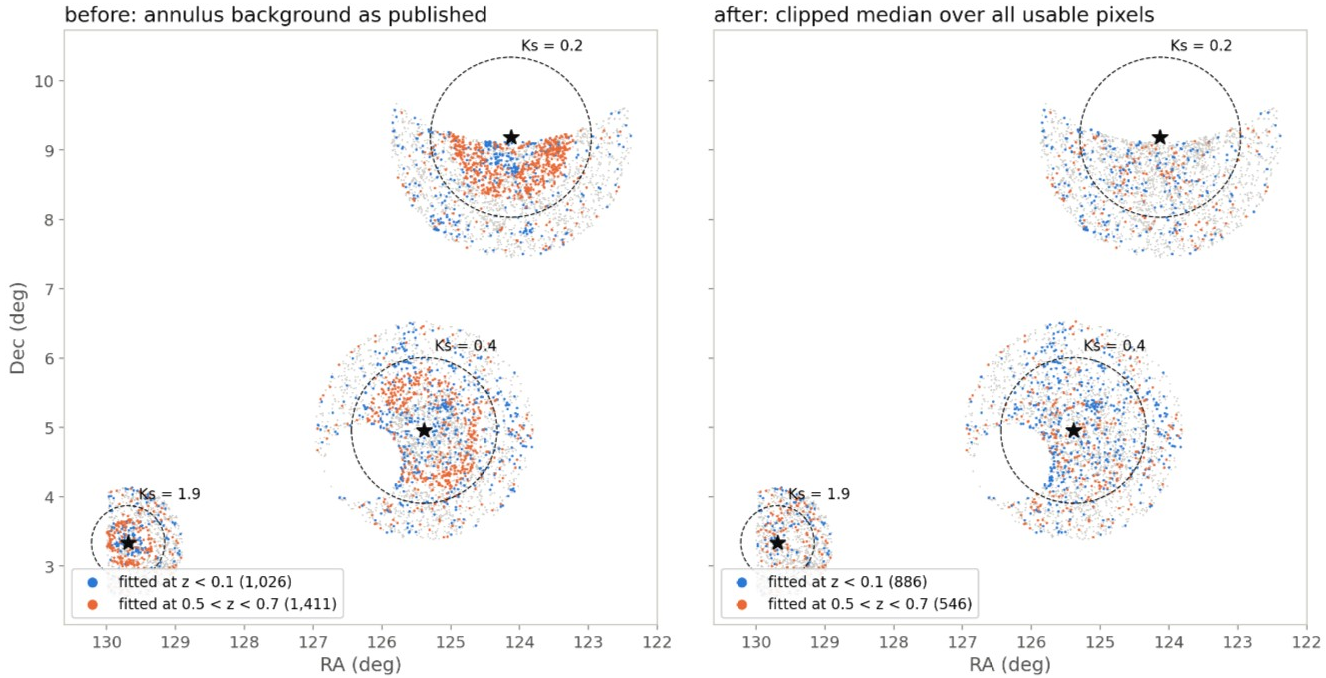
UNIONS × SPHEREx, r < 20.5, n(z) from matched DESI redshifts, against the DES Y6 3×2pt lens bins (arXiv:2601.14559); each scaled to peak 1.

Backup slides

Everything from here on is for questions.

SPHEREx images have too conservative masking around stars

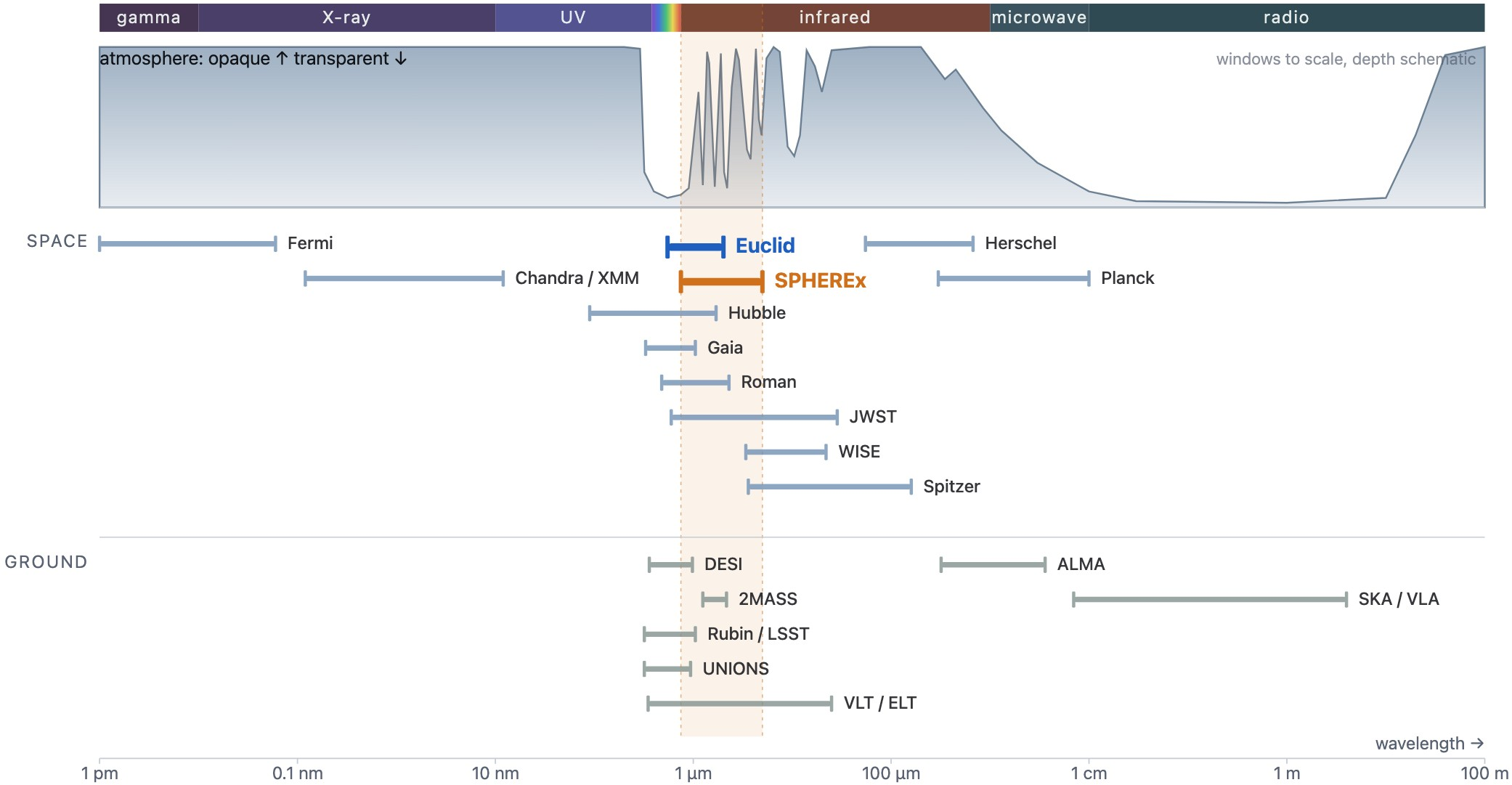




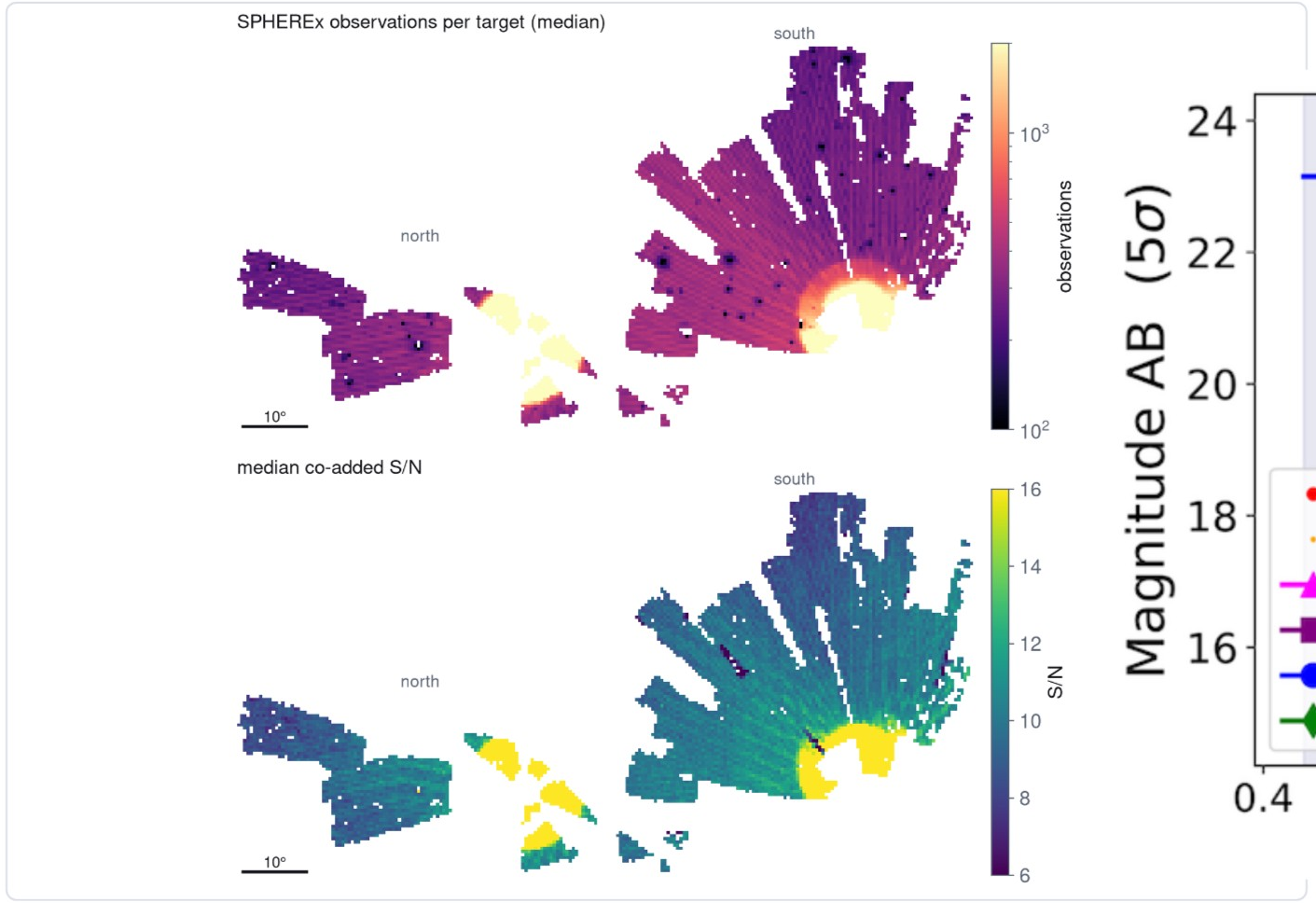
Galaxies round three bright stars: fitted at **z < 0.1** (blue) or **0.5 < z < 0.7** (orange). Left, the annulus background as published; right, the clipped median.

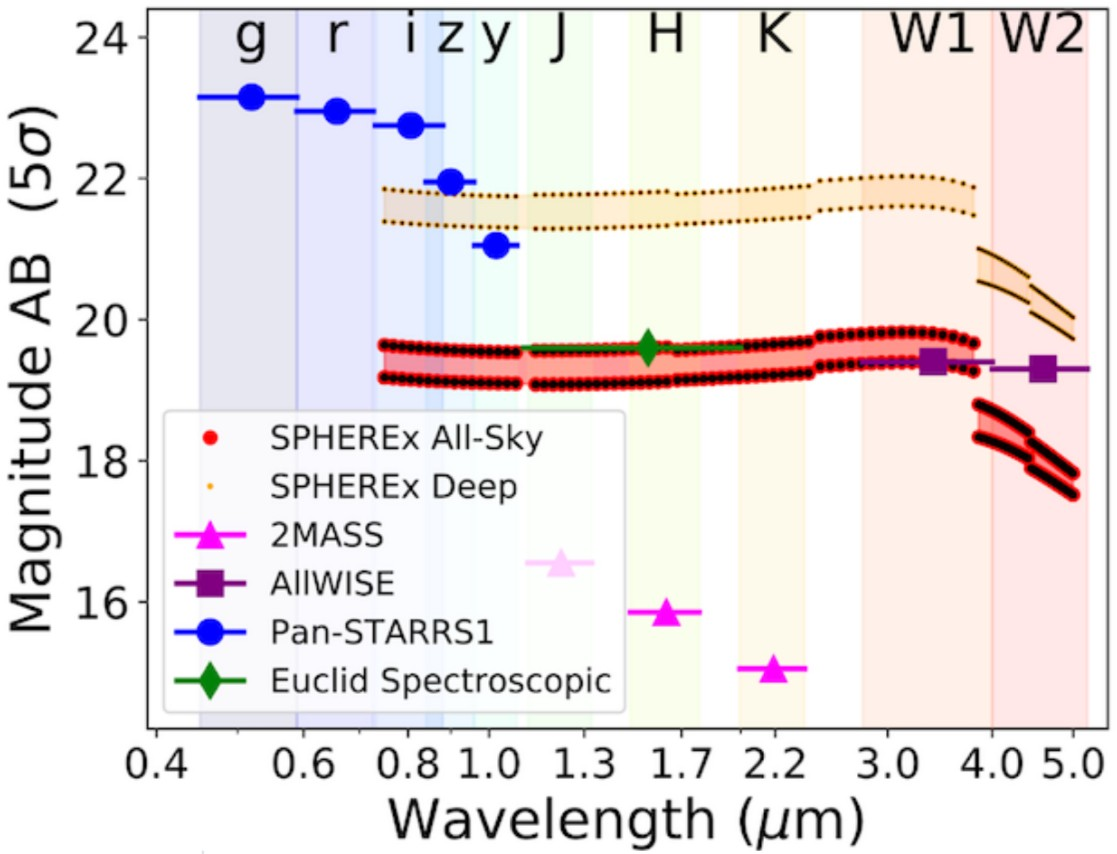
Calum Murray – CEA, Paris-Saclay

The near-infrared has to be observed from space

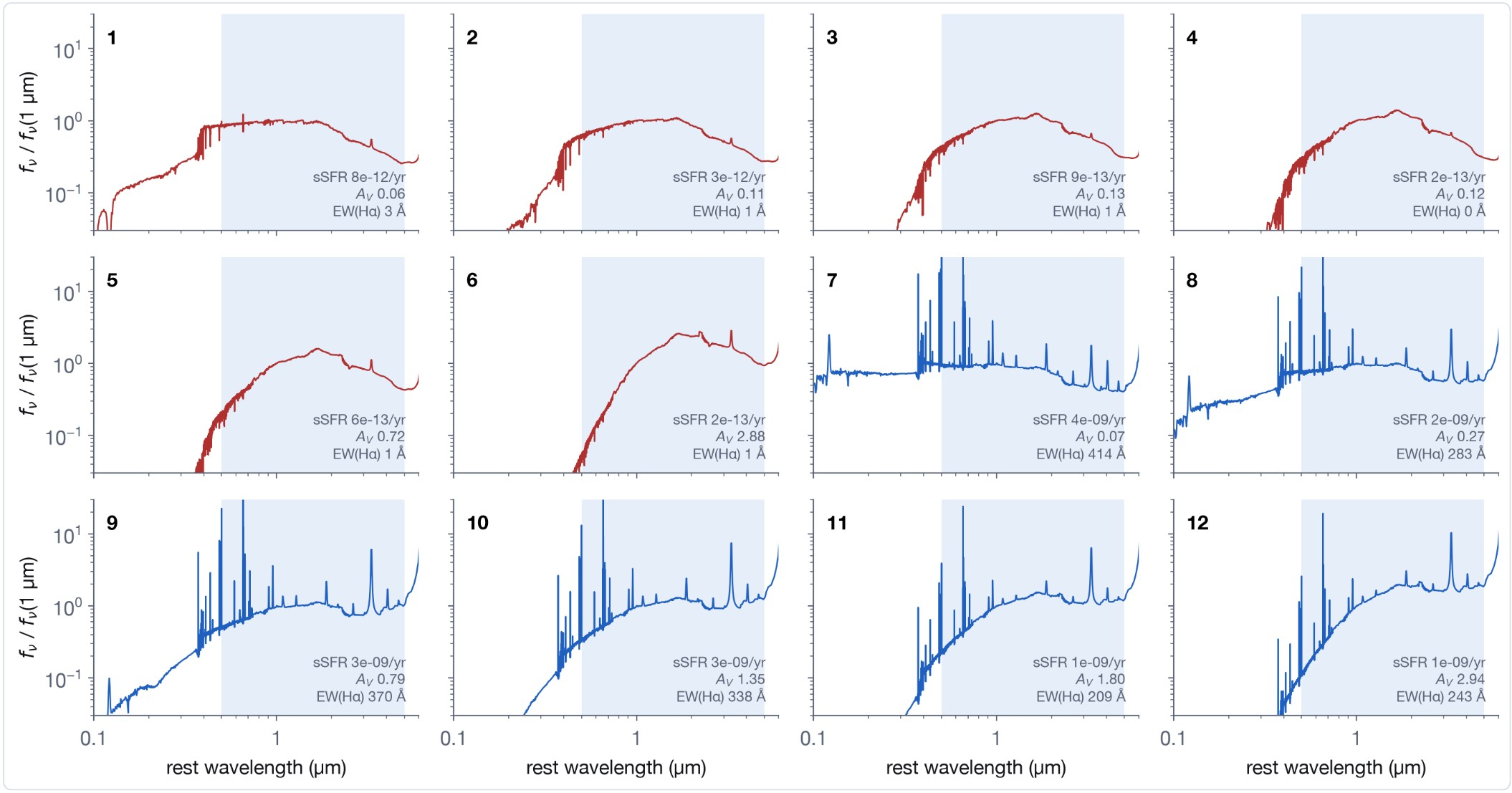


SPHEREx depth

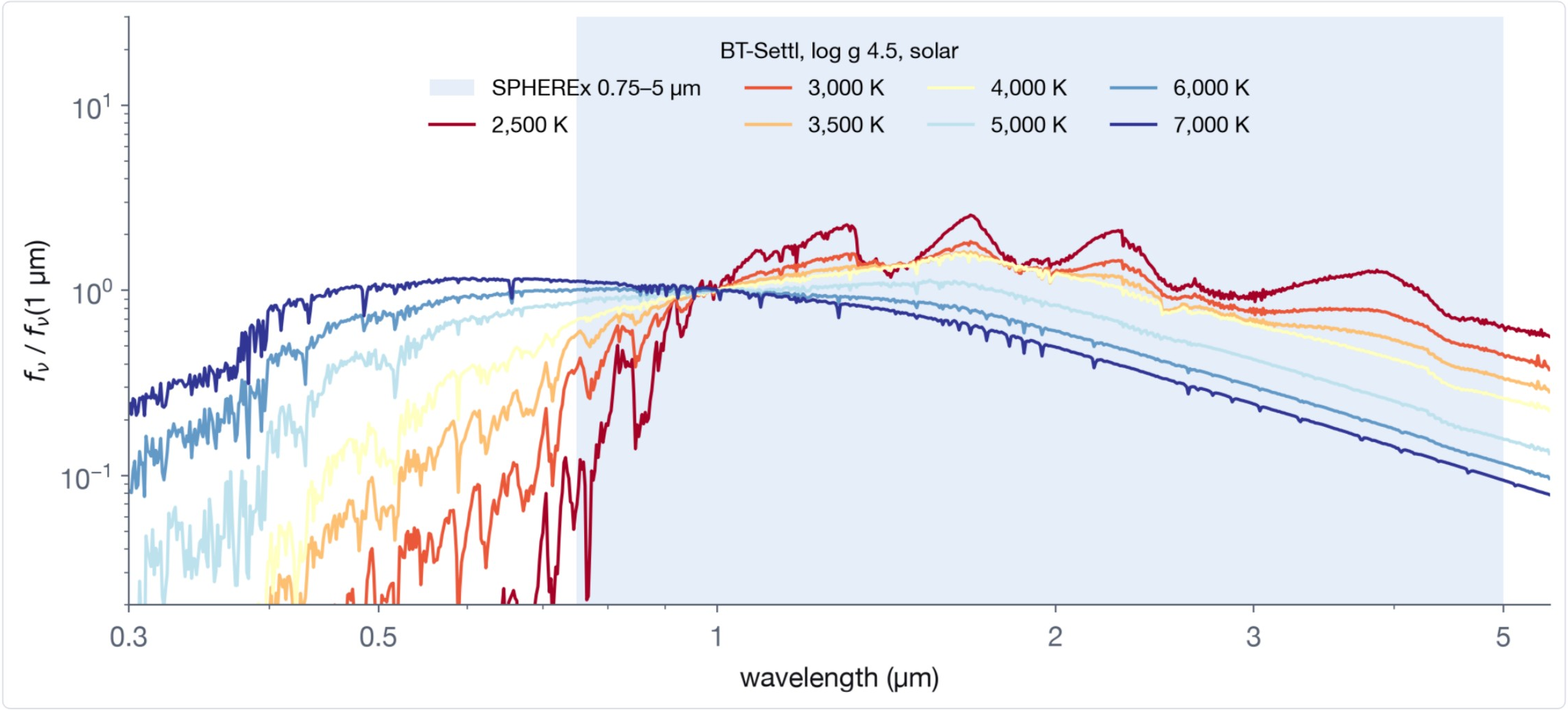




The 12 templates: six quiescent, six star-forming



The stellar grid: BT-Settl model atmospheres, 400–7,000 K



Spectroscopic overlap

CATALOGUE	MATCHES	WHAT IT IS
• DESI DR1	108,622 N 1,559 S	BGS, LRG, ELG, QSO clustering catalogues
• 2dFGRS	6,941	bJ < 19.45, SGP strip + 2° random fields
• 6dFGS	1,049	K < 12.65, whole southern sky, sparse
• OzDES	144	DES supernova fields
2dFLenS	(2,536)	in the report; server down today

